

FIELDCELL · REQUIREMENTS

FieldCell requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/fieldcell		
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REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-02	Amish Chadha	R6 set at 82 kg and the R7 grass pull at 160 N by Amish on 2026-10-02 (FCL-DEC-001); both now met on paper
0.6	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.5	2026-10-01	Amish Chadha	Values and status from FCL-CAL-001 v0.3 for the constructable design (FCL-DDR-003); no target changed
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Status of every requirement from FCL-CAL-001; R2 names 120 V 60 Hz for the first build (FCL-DDR-001); irradiance basis stated
0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

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These are the requirements for the concept. The value and status columns come from the TRL 3 calculation note FCL-CAL-001 v0.3, which checks each one by calculation against the constructable design of FCL-DDR-003. Eight are met, none is not met, one is over its value-engineering target (R12 cost, by \$1), one is at risk (R9) and two cannot be verified until test (R8 and R11). On 2026-10-02 Amish set the R6 mass limit at 82 kg and the R7 grass pull at 160 N, keeping the steel frame and flat-free tyres; the prototype is weighed at TRL 4 and moves to a bolted aluminium frame if it is over 82 kg (FCL-DEC-001); the cost savings worth trying for R12 are in its Value engineering section. On 2026-09-25 Amish accepted the TRL 3 recommendations (FCL-DDR-002): the R6 mass limit is relaxed from 70 kg to 75 kg, the value-engineering target behind R12 moves from \$1,500 to \$2,100, R3 and R4 are stated in global horizontal irradiation, and R8 and R9 name the filter fan and sun shade.

Table 1. Requirements.

ID	Requirement	Target	Value at TRL 3 (FCL-CAL-001 v0.3)	Status	Verification
R1	Store enough energy for a day of field use	1.2 kWh nominal or more; 1.0 kWh usable or more	1.28 kWh nominal, 1.15 kWh usable at 90 % depth of discharge	Met	Calculation; datasheet
R2	Supply AC and DC loads	1,000 W continuous pure sine AC at 120 V 60 Hz for the first build (230 V 50 Hz variant), 2,000 W surge; DC outlets 300 W or more in total (USB-C PD and 12 V)	1 kW inverter by selection; 440 W DC; 66 A continuous battery current on a 100 A fuse and BMS	Met	Calculation; datasheets
R3	Recharge from its own PV	1.1 kWh/day or more into the battery at 4 kWh/m ² /day of global horizontal irradiation (GHI, "peak sun hours"); full recharge from 10 % in one day at 5 kWh/m ² /day GHI with no load	1.17 kWh/day at 4 kWh/m ² /day (1.09 in 40 °C air); full recharge in 0.79 day at 5	Met	PV yield calculation; later logged trial
R4	Carry a reference load through a day without sun	Reference load of 0.9 kWh/day at the outlets for 1 day or more with no sun; energy-neutral at 4 kWh/m ² /day GHI or more	1.08 days; +10 % surplus at 4 kWh/m ² /day; break-even at 3.64	Met	Load and energy balance; later logged trial
R5	Deploy fast with one person	From arrival to power available in 10 min or less, one person, no tools; stow in 10 min or less	6.5 min deploy, 5.5 min stow by task analysis, including the tie bars	Met	Task analysis; later timed trial
R6	Movable by one person	Total mass 82 kg or less (relaxed from 70 kg to 75 kg in FCL-DDR-002 and to 82 kg on 2026-10-02, FCL-DEC-001); vertical handle force 10 to 150 N (never negative) over a 5° pitch either way; heaviest removable module 25 kg or less	80.0 kg; 14 to 54 N; pack 12.0 kg	Met on paper (mass 2.0 kg under; to be weighed at TRL 4)	Mass budget and CG calculation; later weighing
R7	Cross rough ground	16 in (406 mm) wheels or larger; ground clearance 150 mm or more; climb a 150 mm step; pull force 150 N or less on a 10 % grade over gravel and 160 N or less over grass (grass relaxed from 150 N on 2026-10-02, FCL-DEC-001); static lateral tip angle 25° or more	406 mm wheels; 193 mm clearance; step with about 149 N pull at the grip; 125 N gravel, 156 N grass; 32.7°	Met on paper (grass pull 4 N under)	Geometry and force calculation
R8	Work in rain and dust	Electronics enclosure IP54 while operating, ventilated by a thermostat-controlled IP54	IP54 enclosure with IP54 filter fan and exhaust filter;	Not verifiable at TRL 3	

ID	Requirement	Target	Value at TRL 3 (FCL-CAL-001 v0.3)	Status	Verification
		filter fan and exhaust filter (FCL-DDR-002); battery enclosure IP65; all cable entries through glands or sealed connectors	IP65 case; glands, by selection		Design review; later spray and dust test
R9	Stay safe across field temperatures	Operate at -10 to 45 °C ambient; BMS blocks charging below 0 °C cell temperature; inverter derating known at 45 °C; reflective sun shade over both enclosures when deployed (FCL-DDR-002)	Charge block by BMS selection; electronics box about 53 °C at 45 °C ambient with the fan and shade (70 °C sealed, unshaded); pack under the shade about 48 °C at 45 °C ambient, so charging stops only above about 42 °C ambient; derating data not yet obtained	At risk	Thermal estimate; datasheets
R10	Stay put in wind when deployed	Wings and cart stable to 15 m/s (54 km/h) with the four outrigger feet staked or ballasted; stow instruction above that	Staked: 28 N pull-out per foot at 15 m/s (50 N with a 1.5 factor); unstaked wings lift at about 8.9 m/s	Met (staked)	Wind load calculation
R11	Be quiet	45 dB(A) or less at 1 m at 500 W load	About 41 dB(A) from assumed fan levels, not datasheet values	Not verifiable at TRL 3	Datasheets; later sound level measurement
R12	Be electrically safe and affordable	Every circuit fused at its source; GFCI (120 V) or 30 mA RCD (230 V) on all AC outlets; battery disconnect reachable without tools; no inlet or path to building wiring; parts cost at or below the value-engineering target of \$2,100	Safety provisions met by design (class T fuse inside the battery case, within 150 mm of the terminal); parts about \$2,101	Safety provisions met by design; over the value-engineering target by USD 1	Design review; priced BOM

■ Assumptions

- "Peak sun hours" in R3 and R4 is daily global horizontal irradiation in kWh/m²/day, the figure most climate data give. FCL-CAL-001 converts it to the east-west wings with a transposition ratio of about 0.95. Confirmed by Amish, 2026-09-25 (FCL-DDR-002).
- 4 to 5 kWh/m²/day covers clear-season sites at low and middle latitudes; a cloudy week or a high-latitude winter is not covered by R3 or R4.
- The reference load in R4 is defined in FCL-PRC-001, Table 3: lighting, radio and phone charging, a laptop and router, and short power-tool bursts.
- Usable battery capacity is 90 % of nominal, for LiFePO₄ cells with a conservative BMS.

- "Arrival" in R5 means the cart has been wheeled to the chosen spot; choosing the spot is not timed.
- The target in R12 is `budget_usd` in `project.yaml`, a hypothetical control target that keeps the design thinking along a value-engineering lens. It moved from \$1,500 to \$2,100 (BOM plus a small contingency) by Amish's decision of 2026-09-25 (FCL-DDR-002). Supplier quotes for the pack, PV wings and inverter will give the real cost.
- The R6 mass limit of 82 kg (set on 2026-10-02) keeps the decided steel frame and flat-free tyres and leaves a 2 kg margin over the 80.0 kg constructable cart; above 82 kg at weighing the frame becomes bolted aluminium. The cart is a two-person lift either way; the handle force is what one person feels, and it stays within limits. The R7 gross pull limit of 160 N follows the same decision.